WEB TECHNOLOGIES
CSE-4004

LECTURE 5: Objects

By: Dr.MD.Sirajuddin

Object

- An object is a real-world entity having properties and methods.
- An object is a collection of key-value pairs.
- Key is a property name
- Value can be of any type: numeric, string, object, function etc.
- Object is a non-primitive datatype that organizes data in a structured way.

Creating Objects: Object Literal

Example:

```
let Student = {
        name:"XYZ",
       age:20,
      cgpa:9.0
      };
```

Accessing Object Properties

- **Dot Notation**

```
document.write(student.name);
```

- **Bracket Notation**

```
document.write(student["name"]);
```

Object with methods

```
let Student = {
            name: "ABC",
            course: "CSE",
        exam()
        {
            alert(this.name+"Attempted Exam");
        }
        }; Student.exam();
```

Method assigned to Object Property

```
let Student = {
        name: "ABC",
        course: "CSE",
        start: function ()
    {
        alert(this.name+"Attempted Exam");
    }
    };

Student.start();
```

Object Creation using *new & constructor*

```
function student(name,course) // Constructor
{
this.name=name;
this.course=course;
}
let s1=new student("ABC", "CSE");
```

Classes

```
class Student {
  constructor(name, age)
      {
      this.name = name;
      this.age = age;
  }

  display()
      {
      alert("Hello!! " + this.name);
      }
      }
```

- **Classes**

```
let s1=new Student("Bob" , 20);
s1.display();
```

- **Deleting Object Properties**

```
let student = {
  name: "Bob",
  age: 20,
  cgpa: 9.0
};
delete student.age;
console.log(student);
```

Output { name: "Bob", cgpa: 9.0 }

- **Adding Object Properties**

```
let student =
{
  name: "Bob",
  age: 20
};

student.cgpa = 9.0; // properties can be added dynamically
console.log(student);
```

Output: {name: 'Bob', age: 20, cgpa: 9}

- **Object Iteration: for … in loop**

```
let Student =
{
  name: "Bob",
  age: 20
};
for (let key in Student)
{
    console.log(key + ": " + Student[key]);
}
```

Output:

```
name: Bob
age: 20
```

Object Methods	Syntax	Output
Object.keys()	console.log(Object.keys(student));	["name", "age"]
Object.values()	console.log(Object.values(student));	[“Bob", 20]
Object.entries()	console.log(Object.entries(student));	[["name",“Bob"], ["age",20]]

Array of Objects

Using Constructor

```
function Student(name, age)
{
    this.name = name;
    this.age = age;
}
let record = [ ];
record.push(new Student("Bob", 20));
record.push(new Student("Alice", 21));
record.push(new Student("Charlie", 19));
console.log(record[1].name); // "Alice"
```

Built in Objects

Date Object : Handles dates and times

Method	Description	Example
new Date()	Create a new date object	let d = new Date();
getDate()	Day of the month (1–31)	d.getDate(); 24
getMonth()	Month (0–11)	d.getMonth();
getFullYear()	Year	d.getFullYear();
getDay()	Day of the week (0–6)	d.getDay();
toLocaleDateString()	Formatted date string	d.toLocaleDateString(); ‘24/01/2026'

Built in Objects

Math Object : Perform mathematical Operations

Method	Description	Example
Math.round(x)	Round to nearest integer	Math.round(4.7); // 5
Math.floor(x)	Round down	Math.floor(4.7); // 4
Math.ceil(x)	Round up	Math.ceil(4.2); // 5
Math.random()	Random number 0–1	Math.random();
Math.max(a,b,...)	Maximum value	Math.max(3,7,2); //7
Math.sqrt(x)	Square root	Math.sqrt(16); //4

Built in Objects

Document Object : Access and Manipulate HTML DOM

Method	Description	Example
getElementById(id)	Select element by ID	document.getElementById("demo");
querySelector(selector)	Select first matching element	document.querySelector(".class");
createElement(tag)	Create new element	document.createElement("div");
appendChild(node)	Add element to DOM	parent.appendChild(child);
innerHTML	Change element content	elem.innerHTML = "Hello";

Built in Objects

Map Object : *Stores key-value pair (key can be of any type)*

let m= new Map([["name", "Alice"], ["age", 25],]);

Method	Description	Example
set(key,value)	Add/update key-value	m.set("course", "CS");
get(key)	Retrieve value	m.get("a"); //1
has(key)	Check if key exists	m.has("b");
delete(key)	Remove key	m.delete("a");
size	Number of entries	m.size;
clear()	Remove all entries	m.clear();

Built in Objects

Set Object : Stores unique values (no duplicates)

let s = new Set();

Method	Description	Example
add(value)	Add value	s.add(1);
has(value)	Check existence	s.has(2); returns true or false
delete(value)	Remove element	s.delete(1);
clear()	Remove all	s.clear();
size	Number of elements	s.size;
forEach()	Iterate through elements	let s=new Set([1,2,3]); s.forEach(v =>console.log(v));